

Source Audit – Lab 05 – Race Condition

****Resolved lab****: `labs/05-race-condition`

****Series****: Senior Software Engineer Daily #05

****Repo URL****: <https://github.com/lukman-ss/software-engineering-lab/tree/main/labs/05-race-condition>

****Git status****: clean (working tree matches HEAD)

****Files read (human-authored)****:

- README.md
- datarace.go, datarace_test.go, datarace_unsafe_test.go
- atomic_inventory.go, atomic_update_test.go
- unsafe_inventory.go, mock_inventory.go, lost_update_test.go
- inventory.go, booking_test.go
- postgres_rowlock.go, postgres_booking.go
- schema.sql
- go.mod, go.sum

****Supporting files outside lab****: none (all needed code resides in lab directory).

****Excluded / inaccessible****: generated binaries, Docker volumes, external docs.

****Demonstrates****: business-level race conditions (lost update) and safe alternatives (atomic conditional update, row-level lock, UNIQUE constraint).

****Evidence ledger**** (selected examples):

- `TestLostUpdate\_Deterministic` shows invariant break (initial $1 \neq 2 + 0$) – lines 24-33, 146-154 in `lost\_update\_test.go`.
- `TestAtomicUpdate\_StockOne` verifies atomic decrement preserves invariant – lines 42-63, 100-115 in `atomic\_update\_test.go`.
- `Test500\_ConcurrentBooking` proves UNIQUE constraint prevents double booking – lines 61-73, 120-144 in `booking\_test.go`.
- `PostgresRowLockRepository.TrySell` uses `SELECT ... FOR UPDATE` to serialize updates – lines 30-67 in `postgres\_rowlock.go`.
- `AtomicInventory.DecrementStock` implements single-statement conditional update – lines 46-60 in `atomic\_inventory.go`.

****Title candidates****:

1. “Lost Update & Safe Countermeasures in Inventory Systems”
2. “Race Condition pada Penjualan Stok: Dari Bug ke Solusi”
3. “Mengatasi Lost Update dengan Atomic Update & Row Lock”
4. “Invariant Rusak Karena Race Condition – Analisis & Perbaikan”
5. “Concurrency pitfalls in inventory: lost update, booking clash, and fixes”

****Selected title****: #2 – clear, concrete, matches Indonesian language requirement, and reflects actual lab focus.

****English variant****: “Race Condition in Stock Sales: From Bug to Solution”

****Style references****: not accessed.